

JOYDEEP BHATTACHARJEE

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Lead Machine Learning Engineer

Lead Machine Learning Engineer with 14+ years of experience in GenAI, LLMs, and GPU-accelerated inference. Currently leading core GenAI services for Adobe Firefly with PyTorch, CUDA, Triton, and TensorRT. Delivered 70% inference latency reduction via LLM quantization, scaled NLP pipelines to 10M+ messages/day, and published two books in machine learning.

TECHNICAL STRENGTHS

AI / ML :	LLM, Deep Learning, Machine Learning, NLP, Computer Vision
Performance Engineering :	CUDA, Triton, TensorRT, LLM Quantization, Inference Optimization, GPU F
Languages & Frameworks :	Python, PyTorch, TensorFlow, Flask
Infrastructure :	AWS, Azure, GCP, Kubernetes, MySQL, PostgreSQL, Elastic Search, MongoD

EXPERIENCE

Adobe Sep 2025 - Present
Lead Machine Learning Engineer - GenAI, PyTorch, CUDA, Triton, TensorRT, Diffusion Models Bengaluru, India

- Lead technical development of core GenAI services and APIs integrating generative models on Adobe Firefly, architecting enterprise-scale ML workflows for model customization, serving, and ecosystem integration with both Adobe's first-party and third-party generative models.
- Build and optimize GPU-accelerated pipelines for model training and inference using PyTorch, CUDA, Triton, and TensorRT; implement custom CUDA/Triton kernels, kernel fusion, and mixed-precision techniques to maximize tensor core utilization and reduce memory bandwidth bottlenecks for diffusion models and transformers at scale.
- Drive cross-functional alignment with Product Managers, TPMs, and engineering leaders to define and deliver on GenAI Services roadmap; mentor ML engineers in distributed systems, Kubernetes, and GPU resource management.
- Research and evaluate emerging ML and MLOps technologies to enhance engineering velocity and system performance across the organization.

Samsung Semiconductor India Research Jul 2022 - Sep 2025
Senior Machine Learning Staff Engineer - RAG/LLM/NLP/GenAI, tensorflow, pytorch, python Bengaluru, India

- Performance tuning and inference optimisation for state-of-the-art LLM models for in-house NPU computer architecture using quantization and graph optimization techniques.
- Research and implementation of novel Retrieval Augmented Generation (RAG) + LLM application for breakdown maintenance of semiconductor manufacturing equipment. Built application from scratch with precision of 96%. Applied LLM Quantization techniques on llama3 model to bring response time from 30 seconds to less than 10 seconds by 70%.
- Applied Research in the area of Deep Learning and its application in semiconductor manufacturing processes. Responsible for improving FAB yield using Deep Learning and AI models. Work on original research. As part of project performed more than 100 experiments in 2024.
- Handle large datasets and develop data pipelines to provide inputs for training and testing models.
- Lead and mentor a team of AI researchers, define technical roadmap and architecture for AI projects. Prepare and submit presentations and project reports to upper management.
- Collaborate with cross-functional teams, including product managers, and front-end developers, to design solutions that meet business goals.

yellow.ai

Engineering Manager (NLP) - ML, NLP, python, tensorflow, pytorch

Jan 2021 - Jul 2022

Bengaluru, India

- Designed and implemented next-generation AI/NLP infrastructure for deploying thousands of models at scale in multi-lingual, multi-modal conversational AI; led MLOps initiatives and scaled inference pipeline from 4M to 10M messages/day within 1 year.
- Built production-ready NLP pipelines with pre/postprocessing and native Kubernetes scaling using Torch/Transformers/TensorFlow across full message flow stack.
- As engineering lead, owned client issues, architecture decisions, hiring, mentorship, POCs, and cross-functional alignment on new product features.

Nineleaps Technology Solutions

Technical Lead - Data Science, Machine Learning

May 2017 - Dec 2020

Bengaluru, India

- Designed ML and optimization architectures for Inventory Forecasting system across 40,000 products and thousands of stores; built model deployment lifecycle from ground up: versioning, deployment, monitoring, and dashboarding.
- Built public health knowledge graph with ETL pipelines and graph DB querying layer at scale of 1B+ edges and 200M+ nodes.

Previous Experience

Python Developer

2011 - 2017

Bengaluru & Kolkata

- HackerEarth (Category Head - Python, 2016–17) · SLK Software Services (Software Engineer, 2015–16)
- Tata Consultancy Services (Systems Engineer, 2011–15)

EDUCATION

National Institute of Technology, Silchar

B.Tech in Electrical Engineering

GPA: 6.98

June 2007 - July 2011

PUBLICATIONS

Bhattacharjee, Joydeep. Practical machine learning with Rust: Creating intelligent applications in Rust. Apress, 2019.

Bhattacharjee, Joydeep. fastText Quick Start Guide: Get started with Facebook's library for text representation and classification. Packt Publishing Ltd, 2018.

OTHER LINKS

Website :	https://infinite-joy.github.io/
LinkedIn :	https://www.linkedin.com/in/joydeep-bhattacharjee-934a1157/
YouTube :	https://www.youtube.com/@joydeepbhattacharjee3849
Github :	https://github.com/infinite-Joy
HuggingFace :	https://huggingface.co/infinitejoy
Medium :	https://joydeep31415.medium.com/